

ORIGINAL ARTICLE

Identifying drug-resistant epilepsy using Swedish National Patient Registry data

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Abstract

Background: Approximately 30% of individuals with epilepsy are drug-resistant, suffering low quality of life and should be evaluated for other treatment modalities. Drug-resistant epilepsy has no specific diagnostic code, making identification in national healthcare registers challenging and limiting epidemiological research based on register data.

Methods: To define our cohort, we used 1320 individuals from the Swedish National Epilepsy Surgery Register (SNESUR) as index individuals. Each index individual was matched with up to 17 individuals with epilepsy of similar age and diagnosis year in the Swedish National Patient Register. After excluding all SNESUR individuals, the study cohort consisted of 21 393 individuals, a cohort broadly representative of the general epilepsy population. Data on diagnoses, procedures, healthcare contacts, and dispensed antiseizure medications were obtained from national registers. A distance-based ranking algorithm was developed and validated against medical records. Three machine-learning classifiers were then trained and tested using a proxy of the International League Against Epilepsy definition of drug-resistant epilepsy.

Results: The ranking algorithm showed high performance for identifying drug-resistant epilepsy, with a positive predictive value (PPV) of 86.2% (95% CI: 0.81–0.90). Out of the three machine-learning classifiers, Gradient Boosting Machine achieved the highest F1-score of 0.762 (along with PPV = 0.781, sensitivity = 0.744, specificity = 0.820, area under the precision-recall curve = 0.858). The most significant variables found were “total visits to outpatient clinic due to epilepsy,” “age of onset of epilepsy,” “total admissions due to epilepsy,” and “total days of admission due to epilepsy.”

Conclusion: Both the ranking algorithm and machine-learning approaches showed good performance in register-based phenotyping of drug-resistant epilepsy using routinely collected register data, supporting their use in epidemiologic research.

KEYWORDS

artificial intelligence, drug-resistant epilepsy, epidemiology, epilepsy, machine learning

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1 | INTRODUCTION

Epilepsy affects over 70 million people worldwide and about 30% develop drug-resistant epilepsy (DRE). However, most people achieve seizure freedom with initial antiseizure medications (ASM).¹⁻⁴

DRE has no specific diagnostic code in healthcare registers. Consequently, nationwide register data cannot readily be used to identify individuals with DRE, limiting epidemiological studies of its risk factors, healthcare utilization, and long-term outcomes.

According to the International League Against Epilepsy (ILAE), the definition of DRE is defined as treatment failure by two appropriately chosen medications, in adequate and tolerated doses, used either separately or in combination to reach seizure freedom.⁵⁻⁷ Individuals with DRE are disproportionately burdened, as persistent seizures and cumulative ASM side effects cause significant physical, cognitive, and psychosocial challenges, reducing quality of life.⁸⁻¹⁰

It might take years after onset until the epilepsy is considered drug-resistant, a pattern more frequent in childhood and early adolescence onset epilepsy.¹¹ The risk of DRE is influenced by many factors such as early onset, type of epilepsy, multiple seizure types, high seizure frequency, and generalized epileptiform activity on surface.^{12,13} As the risk factors interact in a complex manner, there is an absence of a single overarching biomarker.

In healthcare registers, DRE is characterized by a combination of treatment history, healthcare utilization, hospital admissions, diagnostic investigations, and clinical characteristics. Consequently, identifying individuals with DRE in nationwide register data is challenging using conventional approaches.

Artificial intelligence and machine learning (ML) can detect patterns, enabling integration of multidimensional clinical data to improve phenotypic characterization and risk stratification.¹⁴ ML has previously been used in DRE prediction, identification of surgical candidates, and surgical outcome prediction.¹⁵ However, comparatively few studies have focused on register-based phenotyping of established DRE in nationwide healthcare registers.

The aim of this study was to develop a computational algorithm along with training ML models to identify DRE in the Swedish National Patient Register and to validate the accuracy against medical records.

2 | MATERIALS AND METHODS

This cohort study includes individuals from a database created based on index individuals from the Swedish National Epilepsy Surgery Register (SNESUR), included between 1990 and 2018. For each index individual, up

Key points

- Drug-resistant epilepsy can be reliably identified in the Swedish National Patient Register using both our developed ranking algorithm and a machine-learning classifier using only data from the Swedish National Patient Register.
- The positive predictive value for drug-resistant epilepsy using the ranking algorithm was 86.2%.
- Gradient Boosting Machine achieved an F1-score of 0.762 for drug-resistant epilepsy.

to 17 individuals diagnosed with epilepsy (ICD-9: 345, except 345Q; ICD-10: G40) in the same year and at the approximate same age (the year of birth was initially adjusted by ± 1 year and subsequently by ± 2 years, ± 3 years, ± 4 years and so forth) were identified in the Swedish National Patient Register (SNPR) and included in the cohort.

Data were obtained from the Swedish National Board of Health and Welfare, including inpatient admissions and outpatient visits with information covering epilepsy, comorbidities, and performed examinations. The SNPR is a nationwide administrative healthcare register maintained by the Swedish National Board of Health and Welfare, providing nationwide coverage of inpatient care since 1987 and specialist outpatient care since 2001. The register contains information on diagnoses, healthcare utilization, and performed investigations but does not contain a specific diagnostic code for drug-resistant epilepsy. Prescribed medications were retrieved from the Swedish Prescribed Drug Register. The 1320 SNESUR index individuals were not used in our analysis because they would introduce a selection bias potentially inflating the apparent performance of the ranking algorithm and ML classifiers. The final analytical cohort comprised 21 393 individuals.

2.1 | Distance-based ranking algorithm

Fourteen variables were used including measures of hospital admissions, inpatient days, outpatient visits, multidisciplinary encounters, number of status epilepticus episodes and associated hospitalization days, the number of performed MRIs, CTs, EEGs, and outpatient visits and admissions due to psychiatric disorders.

There were no missing values across the variables, all of which were discrete numerical integers. Because the algorithm relies on distance calculations, min-max normalization was used to scale each variable to the range [0,1], ensuring all variables contributed equally to the analysis.

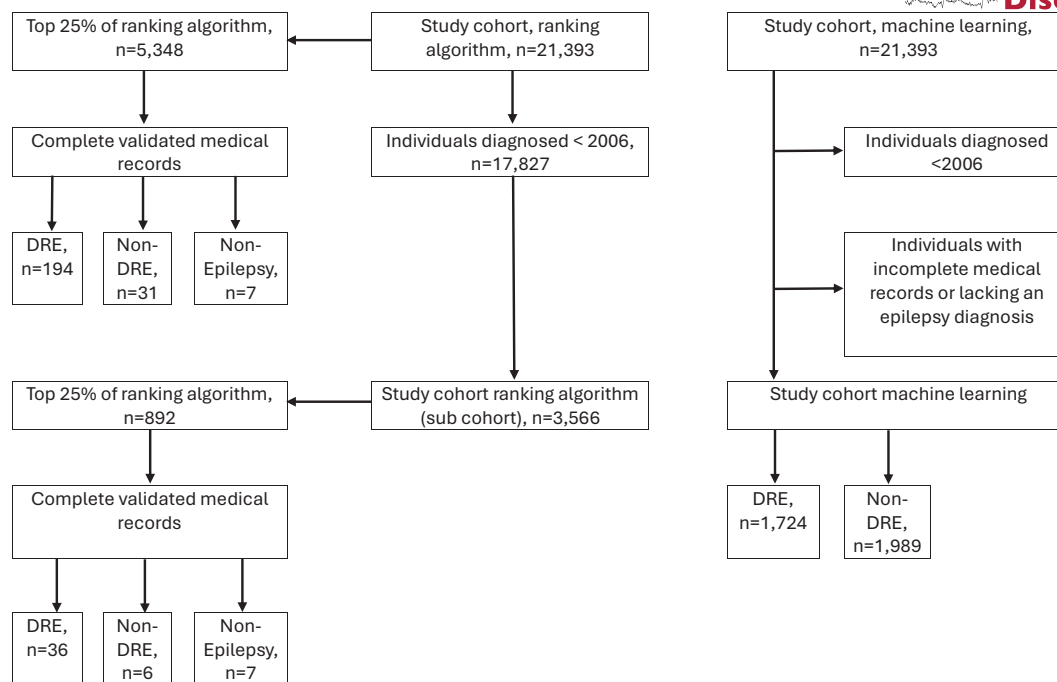


FIGURE 1 Flow chart illustrating patient cohorts selection.

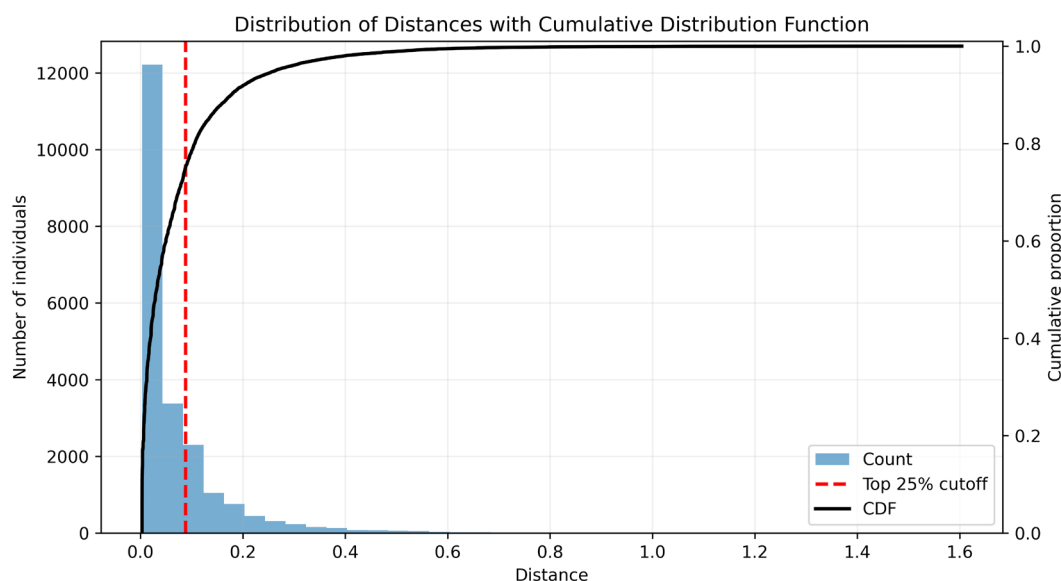


FIGURE 2 Distribution of distance values with corresponding cumulative distribution function (CDF): The histogram (blue) shows frequency, the black curve shows cumulative proportion, and the red dashed line marks the top 25% cutoff.

After normalization, all individuals (Figure 1) were ranked by the likelihood of DRE. Each individual was represented as a vector in a 14-dimensional space (each dimension representing a variable), with rankings based on the vector's magnitude, or the Euclidean distance from the origin. This was motivated by the hypothesis that higher values are associated with greater disease severity and increased likelihood of DRE.

The top 25% of the ranking was selected pragmatically as the primary threshold, acknowledging that although approximately 35% of individuals with epilepsy

are estimated to have DRE, the clinical boundary remains inherently fluid.

To evaluate this cutoff, the distribution of distances was visualized using a histogram with an overlaid cumulative distribution function (CDF), illustrating both the number of individuals across distance ranges and the cumulative proportion of individuals with distances up to a given value (Figure 2).

Sensitivity analyses were also performed using stricter thresholds, including the top 10% and top 1% of the cohort ranking. Positive predictive values (PPVs) were calculated

using clinically validated individuals within each threshold. DRE enrichment was further assessed using two-sided Fisher's exact tests, comparing individuals in the top 10% with those ranked between the 10th and 25th percentiles, and individuals in the top 1% with those ranked between the 1st and 25th percentiles. These analyses were intended to evaluate whether increasing ranking distance was associated with progressively greater DRE enrichment rather than to identify an optimal cutoff. Because clinical validation was restricted to individuals within the original top 25%, comparisons below this threshold and formal cutoff optimization were not feasible.

A subcohort of 3566 individuals (Figure 1) with ASM counts in the Swedish Prescribed Drug Register was created, limited to those diagnosed after December 31, 2005 when the register became available. The ranking algorithm was applied on this subcohort, and PPV was calculated.

The distances in the ranking algorithm were compared against the number of ASMs for each individual, using two types of plots to visualize this relationship.

The scatter plot shows distance (x-axis) versus number of ASMs (y-axis), with lines marking the DRE threshold (≥ 2 ASMs) and the ranking cutoff. A hexbin plot highlights where individuals cluster. Together, they assess whether greater distance is associated with higher treatment burden.

The scatter plot was created to visualize four groups: top 25% with ≥ 2 ASMs (definite DRE), top 25% with < 2 ASMs (probable DRE), bottom 75% with ≥ 2 ASMs (possible non-DRE), and bottom 75% with < 2 ASMs (definite non-DRE). PPV and NPV, with 95% confidence intervals, were calculated for each group using validated cases as the ground truth (Figure 3).

2.2 | Clinical validation

Of the 5348 individuals identified with DRE by the ranking algorithm, 3257 were children and 2091 were adults. Age-group proportions, calculated and ordered by diagnosis date to ensure annual representativeness, were used to select 300 individuals for medical record review and validation. Records were obtained from all treating hospitals for 281 of 300 individuals; 49 cases had insufficient data (Figure 1). Records were independently reviewed by MM, HS, and RW and mapped to predefined variables including sex, age at seizure onset, semiology at onset and currently, seizure frequency, MRI, CT, and EEG results, genetic testing, ASM exposure (count and type of prior medications), history of status epilepticus, and comorbidities.

The PPV was calculated under two definitions of DRE (≥ 2 ASMs and ≥ 3 ASMs), using the validated individuals as ground truth.

2.3 | ML classification

Using the subcohort in combination with the 225 validated individuals, three ML classifiers were evaluated. This cohort consisted of 3713 individuals. A binary target variable was defined, with 1 indicating DRE and 0 indicating non-DRE. All individuals who had tried ≥ 2 ASMs were defined as DRE, and thus, the cohort included 1724 DRE and 1989 non-DRE individuals (Figure 1).

The models were trained twice, once using ≥ 2 ASMs and once using ≥ 3 ASMs. The ML models used 31 variables for training, including 17 comorbidity- and age-related features to provide a comprehensive clinical profile.

Three classifiers were trained: Random Forest (RF), Gradient Boosting Machine (GBM), and Extreme Gradient Boosting (XGBoost), all based on ensembles of decision trees. A decision tree splits data into smaller groups using variable-based rules,¹⁶ although single trees are often weaker. Ensemble methods improve performance by combining many trees. RF uses bagging, training independent trees on random samples in parallel, and aggregating predictions by majority vote. GBM uses boosting, training trees sequentially so each learns from prior errors.¹⁶ XGBoost is a more efficient, optimized implementation of GBM.¹⁷

Precision (PPV), recall (sensitivity), F1-score, specificity, area under the precision-recall curve (AUPRC), and area under the receiver operating characteristic curve (AUROC) were applied to evaluate the models using stratified 5-fold cross validation. AUPRC was included due to imbalanced target class as AUROC alone may provide an overly optimistic assessment of the performance. To further address class imbalance, class weighting strategies were applied: RF used built-in class weighting, while GBM and XGBoost applied fold-specific sample weights. Given the trade-off between precision and recall, and the clinical relevance of both, the F1-score was prioritized as a balanced performance metric.

All analyses were performed for ≥ 2 ASMs and ≥ 3 ASMs, using Jupyter Notebook and Python 3.8.18. Data handling was conducted using pandas 2.0.3 and NumPy 1.24.3, while visualizations were produced using Matplotlib 3.7.2 and seaborn 0.12.2. The RF and GBM models were implemented using RandomForestClassifier and GradientBoostingClassifier, respectively, from scikit-learn 1.3.2. XGBoost was implemented using XGBClassifier from XGBoost 2.1.4. Confidence intervals were calculated using EpiTools.

No data-driven hyperparameter tuning was performed. Instead, model hyperparameters were specified before training, with package-default values used where no alternative settings were defined. Within each cross-validation fold, the models were then fitted to the training data using these

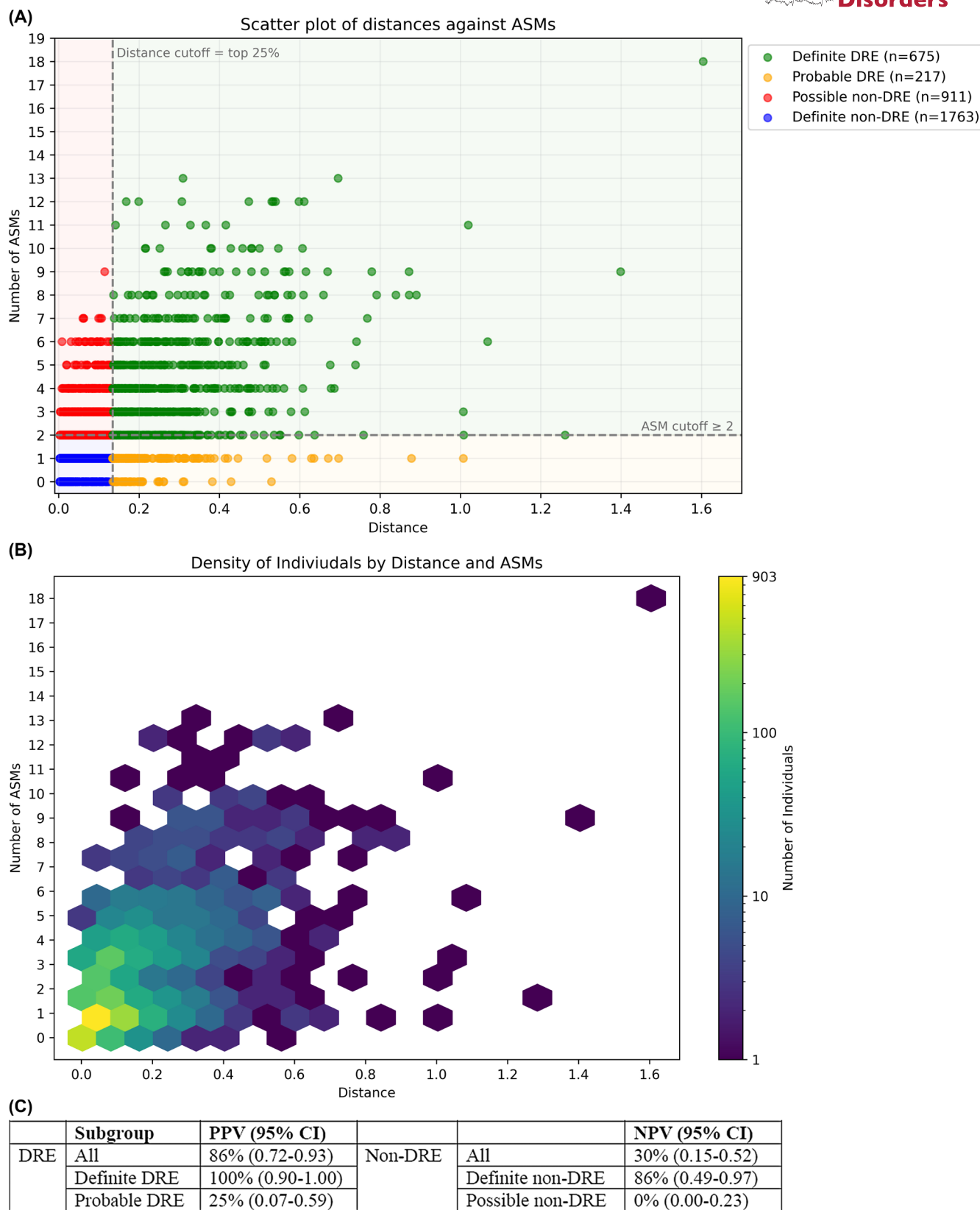


FIGURE 3 Defined groups according to ranking algorithm combined with number of antiseizure medications. (A) Scatter plot of ranking distances against the number of ASMs. The vertical dashed line indicates the top 25% distance cutoff (to the right), and the horizontal dashed line marks the DRE threshold (≥ 2 ASMs). Individuals are colored according to their identification based on these two criteria. (B) Hexbin plot showing the density of individuals across distance values and number of ASMs. Color intensity represents the number of individuals within each bin. (C) PPV and NPV for DRE and non-DRE subgroups, presented with 95% confidence intervals.

hyperparameter settings. For RF, class weight was set to “balanced.” GBM used default hyperparameter values, whereas XGBoost was configured with `objective=“binary:logistic”` and `eval_metric=“logloss.”` Furthermore, for GBM and XGBoost, the balanced sample weights were calculated separately within each training fold and supplied during model fitting. A random seed of 42 was used for reproducibility. In addition, the three classifiers were trained and evaluated using stratified fivefold cross-validation implemented with scikit-learn's StratifiedKFold. A standard rather than nested cross-validation was used because no hyperparameter tuning or internal model-selection procedure was performed that would necessitate an inner selection loop.

The performance metrics were summarized as mean $\pm$ standard deviation across the five stratified cross-validation folds. As a supplementary analysis, two-sided paired DeLong tests were used to assess whether the classifier with the highest AUROC performed statistically significantly better than the other classifiers. To compare AUROC between classifiers, one out-of-fold predicted probability was obtained for each individual from the fold in which that individual was held out. Identical fold assignments were used for all classifiers. The DeLong tests were implemented using the MLstatkit 0.1.91 library.

Finally, to identify the most predictive features, permutation importance was calculated for the overall better-performing classifier, where the top 10 features were ranked according to the mean decrease in AUPRC across the held-out cross-validation folds.

2.4 | Ethical approval

The study was approved by the Swedish Ethical Review Authority (DNR 2021-06335-01; 2022-05814-02). The study did not involve any contact with participants. Informed consent was not required as this is a retrospective study solely including anonymized data.¹⁸

3 | RESULTS

3.1 | Distance-based ranking algorithm

The distribution of the normalized distances, corresponding CDF, and the 25% cutoff are shown in Figure 2.

The majority of individuals had distances near zero, with only a small proportion exhibiting higher values (Figure 2). This is represented by the CDF, showing how quickly the cohort accumulates across the distance range, while the top 25% cutoff consists of a small group with relatively elevated distances. Within this group, 225 individuals were manually validated, yielding a PPV of 86.2%

(95% CI: 0.81–0.90) for identifying DRE using its definition of ≥ 2 ASMs. Additional sensitivity analyses showed a PPV of 96.8% for the top 10% (95% CI: 0.91–0.99), and 100% for the top 1% (95% CI: 0.81–1.00). Among validated individuals in the top decile, 90 of 93 had DRE and among individuals ranked between the 10th and 25th percentile, the corresponding numbers were 104 of 132. Comparison of these groups yielded an OR of 8.08; 95% CI: 2.38–27.46 (two-sided Fisher's exact $p < 0.001$).

The top percentile comprised 16 individuals, all of whom had DRE (100%), and among individuals ranked between the 1st and 25th percentile 178 of 209 were validated having DRE. This difference was not statistically significant (two-sided Fisher's exact $p = 0.137$) and should be interpreted cautiously because of the small group size and the absence of non-DRE cases. Because all individuals in the highest percentile had DRE, resulting in complete separation, an OR could not be estimated.

In comparison, when DRE was defined as ≥ 3 ASMs for the validated individuals, the PPV reduced to 66.8% (95% CI: 0.60–0.73).

Limiting our cohort to only individuals diagnosed after 2005 as a subanalysis, the ranking algorithm identified the top 25% ($n = 892$) as DRE. Using the 42 individuals with available records, a PPV of 85.7% (95% CI: 0.72–0.93) was obtained.

The relationship between ranking distances and the number of tried ASMs per individual is summarized in Figure 3. The scatter plot (Figure 3) illustrates the four subgroups defined by the 25% distance cutoff and ≥ 2 ASM threshold, highlighting general agreement (higher distance relates positively to higher tried ASMs) with some discordance. Figure 3 highlights a strong clustering of individuals at low distance values and 0 or 1 ASMs, with relatively few individuals at higher values.

Comparing the four groups against the validated individuals, Figure 3 shows high PPV for the definite DRE group (100%, 95% CI: 0.90–1.00) and high NPV for the definite non-DRE group (86%, 95% CI: 0.49–0.97), while the intermediate groups demonstrate lower values, reflecting much greater uncertainty.

3.2 | ML classification

The ML classifiers were developed using the sub-cohort, including the 225 validated DRE and non-DRE individuals ($n = 3713$).

All three classifiers demonstrated good overall performance in predicting DRE (see Table 1). GBM classifier resulted in the highest F1-score of 0.762. Its precision (PPV) is 0.781, recall (sensitivity) is 0.744, specificity is 0.820, AUPRC is 0.858, and the AUROC is 0.865.

TABLE 1 Performance of Random Forest, Gradient Boosting Machine, and Extreme Gradient Boosting using stratified fivefold cross-validation with ≥ 2 and ≥ 3 antiseizure medications as definitions of DRE.

Model	Class	Metric	2 ASM or more	3 ASM or more
			Held-out folds mean $\pm$ SD	Held-out folds mean $\pm$ SD
Random Forest (RF)	DRE	Precision (PPV)	0.747 $\pm$ 0.011	0.728 $\pm$ 0.041
		Recall (Sensitivity)	0.731 $\pm$ 0.013	0.627 $\pm$ 0.020
		F1-Score	0.739 $\pm$ 0.010	0.674 $\pm$ 0.027
		AUPRC	0.825 $\pm$ 0.016	0.761 $\pm$ 0.023
	Non-DRE	Specificity	0.785 $\pm$ 0.012	0.913 $\pm$ 0.017
	Overall	AUROC	0.835 $\pm$ 0.011	0.871 $\pm$ 0.011
Gradient Boosting Machine (GBM)	DRE	Precision (PPV)	0.781 $\pm$ 0.009	0.631 $\pm$ 0.029
		Recall (sensitivity)	0.744 $\pm$ 0.020	0.777 $\pm$ 0.031
		F1-Score	0.762 $\pm$ 0.014	0.696 $\pm$ 0.026
		AUPRC	0.858 $\pm$ 0.017	0.788 $\pm$ 0.025
	Non-DRE	Specificity	0.820 $\pm$ 0.007	0.833 $\pm$ 0.019
	Overall	AUROC	0.865 $\pm$ 0.011	0.891 $\pm$ 0.013
Extreme Gradient Boosting (XGBoost)	DRE	Precision (PPV)	0.742 $\pm$ 0.020	0.643 $\pm$ 0.026
		Recall (sensitivity)	0.730 $\pm$ 0.028	0.701 $\pm$ 0.024
		F1-Score	0.736 $\pm$ 0.024	0.671 $\pm$ 0.021
		AUPRC	0.836 $\pm$ 0.020	0.759 $\pm$ 0.028
	Non-DRE	Specificity	0.780 $\pm$ 0.016	0.856 $\pm$ 0.016
	Overall	AUROC	0.842 $\pm$ 0.016	0.863 $\pm$ 0.012

Note: Precision, recall, F1-score, AUPRC (DRE), specificity (non-DRE), and AUROC are reported across the held-out folds.

Abbreviations: ASM, antiseizure medications; AUPRC, area under the precision-recall curve; AUROC, area under the receiver operating characteristic curve; DRE, drug-resistant epilepsy.

GBM has the highest AUROC compared to RF and XGBoost, for both definitions of DRE. To test whether this is statistically significant, two-sided paired DeLong tests were conducted between GBM and RF, and between GBM and XGBoost. For the DRE definition of ≥ 2 ASMs, GBM achieved a significantly higher AUROC than RF (0.864 vs. 0.835; difference: 0.029; DeLong $p = 0.001$) and XGBoost (0.864 vs. 0.842; difference: 0.022; DeLong $p \leq 0.001$).

For ≥ 3 ASMs, all classifiers showed good overall performance, yet lower compared to ≥ 2 ASMs. Similar to the previous definition (≥ 2 ASM), GBM obtained the highest F1-score of 0.696 on the test set (see Table 1). Additionally, GBM achieved a significantly higher AUROC than Random Forest (0.890 vs. 0.871; difference: 0.019; two-sided DeLong $p \leq 0.001$) and XGBoost (0.890 vs. 0.863; difference: 0.026; two-sided DeLong $p \leq 0.001$).

3.3 | Feature importance

Top predictors of DRE include outpatient visits, epilepsy-related admissions and hospital days, age at onset, and,

to a lesser extent, number of video EEGs and intellectual disability (Figure 4).

4 | DISCUSSION

Our distance-based ranking algorithm achieved high accuracy (PPV, 86.2%) for DRE defined as ≥ 2 ASMs. Although this threshold does not fully capture the ILAE definition, it reflects treatment failure and need for consideration of nonpharmacologic interventions.¹⁹ Anchored in this definition, our model showed slightly better performance than prior register-based algorithms using similar two-ASM thresholds; one study reported PPVs of approximately 59% and an F1-score around 0.63.²⁰ Our findings suggest that DRE can be identified with high precision from nationwide healthcare registers despite the absence of a specific diagnostic code. Rather than relying on a single proxy, the algorithm integrates multiple aspects of healthcare utilization and treatment history that together characterize pharmacoresistance. The importance of healthcare utilization variables supports the hypothesis

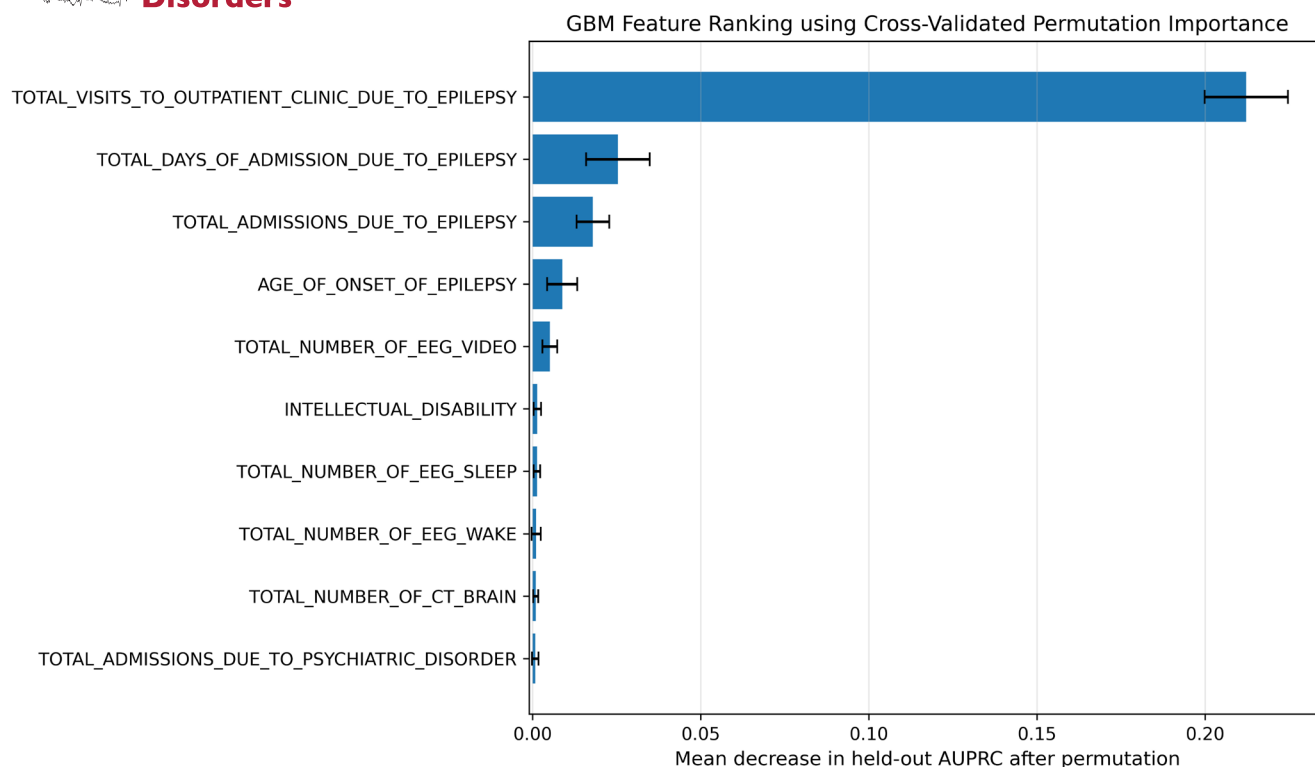


FIGURE 4 Top predictors of DRE include outpatient visits, epilepsy-related admissions and hospital days, age at onset, and, to a lesser extent, number of video EEGs and intellectual disability.

that DRE leaves a multidimensional signature in routine healthcare registers.

Our sensitivity analyses showed that PPV increased to 96.77% for the top 10% and to 100% for the top 1%, suggesting that the distance score reflects the underlying likelihood of DRE rather than providing only a binary classification. We also evaluated the algorithm using a stricter definition of DRE based on ≥ 3 ASMs, which reduced the PPV to 66.8%. In comparison, Hill et al.²¹ validated DRE identification through medical records and reported a PPV of 61% (F1-score 0.710), although their cohort included some individuals without epilepsy. The ≥ 3 ASMs threshold used as a proxy for DRE in previous register-based studies may identify a more narrow cohort. However, it may be used as a proxy when data are limited, considering the broader ILAE definition.

The comparable performance in the 2006–2018 subcohort indicates that the ranking algorithm remained robust across different calendar periods, despite changes in anti-seizure medication use and clinical management over time.

Ranking distance was positively associated with ASM count, suggesting that the algorithm also captures increasing treatment complexity. Agreement with clinical classification was greatest at the extremes of the ranking, where individuals classified as definite DRE or definite non-DRE showed high predictive values. In contrast, individuals with intermediate ranking distances were more heterogeneous

and showed lower predictive values, indicating that the ranking score is less discriminative in borderline cases. These findings suggest that the ranking distance is most useful for identifying groups enriched for DRE in register-based studies rather than for classifying individuals with intermediate scores. This pattern indicates that the ranking score is most informative for identifying individuals who are clearly drug resistant or clearly nondrug resistant.

Among the evaluated ML classifiers, GBM achieved the best overall performance for identifying DRE using the ≥ 2 ASM definition. Although differences in study populations, outcome definitions, and predictor variables limit direct comparisons, our findings are broadly consistent with previous register-based studies demonstrating that ML can identify DRE with good discriminatory performance in register-based data.^{21,22} Compared with previous register-based studies, the higher F1-score observed in our study may reflect the ability of multidimensional register data to improve the balance between precision and recall.

These findings are consistent with the growing body of evidence supporting the application of ML to epilepsy, including prediction of treatment response, pharmacoresistance, and surgical outcomes.¹⁴ More specifically, our results are broadly consistent with previous register-based studies demonstrating that ML can identify DRE with good discriminatory performance.^{21,22}

Applying a stricter definition of DRE based on ≥ 3 ASMs reduced overall classification performance despite improved sensitivity and specificity, illustrating the trade-off between case capture and positive predictive accuracy. Similar findings have been reported in a Danish nationwide register study using a ≥ 3 ASM definition.²³ Together, these findings suggest that ML approaches achieve comparable performance across large population-based registers, while the choice of DRE definition substantially influences model performance.

In our study, frequent outpatient visits were the strongest predictor of DRE, reflecting persistent seizures and repeated medication adjustments. Previous studies align with our findings, showing that closer clinical monitoring and epilepsy-related hospitalizations can serve as practical markers of pharmacoresistance in real-world settings²⁴ and that age at onset is an important risk factor.^{10,25,26}

Clinical variables such as psychiatric comorbidities, including intellectual disability, have been reported as risk factors for DRE.^{27,28} However, in our models, these factors contributed less.

Although no single variable uniquely identifies DRE, combining several complementary indicators appears sufficient to distinguish individuals with pharmacoresistance. The number of ASMs may not capture treatment failure, adherence, dosing, or clinical decision-making, and defining DRE solely by two failed trials may delay recognition. Focusing on observable disease burden and health care utilization may better reflect how pharmacoresistance is identified in routine practice.^{29,30}

4.1 | Strengths

This study has several strengths. First, the use of nationwide registry data, with comprehensive longitudinal coverage and near-complete population representation, enables evaluation of DRE identification in a large, population-based cohort.^{31,32}

Second, algorithm performance was validated against detailed medical records rather than register-based proxy outcomes alone, providing an independent assessment of accuracy. Third, we evaluated both a distance-based ranking algorithm and ML classifiers, allowing a complementary assessment of register-based phenotyping.

4.2 | Limitations

This study has several limitations. First, medical record validation included fewer individuals than planned because not all requested records were available, resulting in less precise performance estimates. Second, the ML

models were trained using a register-derived definition of DRE based on dispensed ASMs rather than clinically confirmed DRE.

Another limitation was that epilepsy diagnoses were identified through the SNPR, although it captured most specialist care in Sweden. Previous validation studies have demonstrated a relatively high validity of epilepsy diagnoses, with a PPV ranging from 79% to 95%.^{33–35}

Similar findings have been reported in Denmark, where a comparable register-based study showed a PPV of 81% for epilepsy.³⁶ This supports the overall reliability of register-based epilepsy diagnoses.

Although registry data provide robust longitudinal coverage, they lack detailed clinical information, including seizure patterns, lifestyle factors, and patient-reported outcomes.

4.3 | Implications of results

Despite these limitations, the proposed approach could provide a practical tool for identifying DRE in nationwide healthcare registers, facilitating large-scale case ascertainment for research and healthcare planning where no dedicated diagnostic code exists.

5 | CONCLUSION

This study demonstrates that DRE can be identified from nationwide healthcare registers using both a distance-based ranking algorithm with a PPV of 86.2% and ML classifiers, despite the absence of a dedicated diagnostic code. By leveraging routinely collected register data, these approaches provide a practical framework for large-scale identification of DRE without additional data collection. The findings support the potential use of registry-based methods in epidemiologic studies. Overall, studies based on registry data may contribute to improved understanding and management of DRE.

AUTHOR CONTRIBUTIONS

Matilda Modigh: Conceptualization, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Software, Validation, Visualization, Writing – Original Draft Preparation, Writing – Review & Editing. **Elina Tayebi:** Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – Original Draft Preparation, Writing – Review and Editing. **Ronny Wickström:** Conceptualization, Funding Acquisition, Resources, Supervision, Validation, Visualization,

Writing—Review and Editing. **Cecilia Bartholdson:** Conceptualization, Supervision, Visualization, Writing – Review and Editing. **Arvind Kumar:** Methodology, Resources, Software, Supervision, Visualization, Writing—Review and Editing. **Heléne Sundelin:** Conceptualization, Data Curation, Formal Analysis, Funding Acquisition (lead), Investigation, Methodology, Project Administration (lead), Resources, Software, Supervision (lead), Validation, Visualization, Writing – Original Draft Preparation, Writing—Review and Editing.

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CONFLICT OF INTEREST STATEMENT

Ronny Wickström has received honoraria for serving on advisory boards for UCB, Jazz Pharma, and GW Pharma and speaker's fees from EISAI, UCB, Sobi, Jazz Pharma, and Sanofi-Genzyme. Heléne Sundelin has received speaker's fees from EISAI. The remaining authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The legal frameworks of Swedish health registers prohibit us from sharing datasets. The Swedish National Board of Health and Welfare and Statistics Sweden have access to all the study data and can be contacted to request the data, following ethical approval from the Swedish Ethical Review Authority. The programming code can be required through the corresponding author.

PATIENT CONSENT

Informed consent was not required because this was a retrospective register-based study using anonymized data and involved no contact with study participants.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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Test yourself

1. What was the main purpose of the distance-based ranking algorithm in this study?
 - A. To predict surgical outcomes in epilepsy
 - B. To classify epilepsy types based on EEG patterns
 - C. To identify individuals with drug-resistant epilepsy (DRE) using registry data
 - D. To determine optimal antiseizure medication dosing
2. Which machine learning model demonstrated the best overall performance in identifying drug-resistant epilepsy?
 - A. Random Forest
 - B. Gradient Boosting Machine (GBM)
 - C. Logistic Regression
 - D. Support Vector Machine
3. Which variable was identified as the most significant predictor of drug-resistant epilepsy?
 - A. Sex
 - B. Number of EEG examinations
 - C. Total outpatient visits due to epilepsy
 - D. Genetic testing results.

Answers may be found in the [Supporting Information](#).